

Primitive-Support Collapse and the Analytic Completion Gate: Tail-Free Abel Certificates, Rectangle Charges, and Physical Extension Invariance

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Abstract

TPC-70 and TPC-71 reduce the first signed cross-key coefficient of the missing-native Gram to a Möbius sum on primitive cofactor coordinates. The general divisor-lifted certificate of TPC-71 allows an arbitrary finitely supported plane kernel and consequently contains diagonal dilations, restricted Mertens sums, and an exact large-diagonal tail. The literal physical kernel has more structure. Its common content is defined canonically by $q = (S_m, S_n)$, and its remaining coordinates $u = S_m/q$, $v = S_n/q$ are coprime. We prove that the support-faithful zero completion is therefore supported on the primitive lattice. Every diagonal dilation $K^{[d]}(x, y) = K(dx, dy)$ with $d \geq 2$ vanishes identically, the exact diagonal tail is zero for every cutoff $D \geq 1$, and the physical divisor lift collapses to ordinary two-dimensional Abel summation at $d = 1$.

For one anchor $\mathfrak{a}$, with $M(x) = \sum_{n \leq x} \mu(n)$, the exact formula is

$$\mathfrak{X}_{\mathfrak{a}} = \mu(a_{\mathfrak{a}})\mu(b_{\mathfrak{a}}) \sum_{x, y \geq 1} M(x)M(y)\Delta_{12}K_{\mathfrak{a}}^0(x, y).$$

We also aggregate all row pairs and anchors before taking absolute values, obtaining a scalar trace kernel and a sharper weighted-plaquette certificate for $\text{tr}(ER)$. An ℓ^1 -valued version retains every anchor and shows that a global cost estimate, rather than uniform anchorwise little-oh notation, is sufficient for trace-level summability.

The tail collapse does not make the variation problem disappear. We prove that mixed variation is exactly the coefficient mass in a unique terminal-rectangle decomposition and that diagonal downsampling contracts this norm. More importantly, the physical sum depends only on values at primitive sites. Any analytic completion at nonprimitive sites gives the same physical sum but may trade a smaller $d = 1$ variation against nonzero higher diagonal slices and tail. We define the resulting completion gauge and prove its exact invariance and certificate. The canonical completion is tail-free, but it need not minimize its $d = 1$ mixed variation within its completion class. Elementary examples separate tail freedom, low variation, signed cancellation, anchor summation, and operator control. Thus the auxiliary diagonal-tail gate for the support-faithful zero extension is closed exactly, while the mass-normalized completion and weighted-plaquette gates remain open. No operator saving, fixed-2 estimate, parity-breaking statement, or prime-pair conclusion is drawn.

Keywords: Möbius function; primitive cofactor plane; canonical zero completion; two-dimensional Abel summation; mixed variation; rectangle atoms; fixed shift; analytic completion.

Convention. Every physical specialization retains one fixed nonzero shift h_0 , literal coefficients, complete native keys, the actual active support, and raw atomic normalization. No statement is specialized to $h_0 = 2$.

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1 Introduction

The primitive-plane route began with a four-corner coefficient in the first mixed trace. TPC-69 shows that the first surviving signed term is cross-key and that its residual sign is carried by the odd erasure

boundary [3]. TPC-70 then gives exact cofactor coordinates: after extracting the common erasure content q , the surviving sign is a product $\mu(a)\mu(b)\mu(u)\mu(v)$ that is frozen along each affine prime ray [4]. TPC-71 applies coprimality inversion and two-dimensional Abel summation to a general finitely supported plane kernel [5]. Its master certificate contains

$$\sum_{d \leq D} \mathfrak{M}_d(U/d) \mathfrak{M}_d(V/d) \mathcal{V}_{\square}(K^{[d]}) + \mathcal{T}_D(K).$$

TPC-72 carries the mass-normalized variation, the representation-dependent diagonal tail before a completion is fixed, and complete anchor summation as open gates; it also warns that a four-corner trace coefficient is not a two-corner erasure entry [6].

The present paper re-examines only the literal support of the kernel before trying to estimate it. The physical coordinates are not an arbitrary pair followed by a coprimality mask. They are defined by

$$q = (S_m, S_n), \quad S_m = qu, \quad S_n = qv.$$

Consequently $(u, v) = 1$ identically. If a kernel is extended by zero at coordinates which do not occur physically, then it is already supported on the primitive lattice. For $d \geq 2$, the point (dx, dy) cannot be primitive. Hence every higher diagonal slice vanishes.

This observation has two different consequences which must not be conflated.

- (i) The support-faithful physical completion is tail-free. Its exact Abel certificate has one slice and needs no growing- d restricted-Mertens theorem.
- (ii) Values away from the primitive lattice are invisible to the physical signed sum. They may therefore be introduced as an analytic completion. Such a completion can fill hard coprimality holes and reduce variation, but its higher diagonal slices and tail return and must be charged.

Thus variation and diagonal tail are not independent invariants of the primitive physical datum. They form a coupled analytic completion gauge. Tail freedom is canonical for the support-faithful zero extension; low variation is a separate theorem.

1.1 Main results

The paper proves seven groups of statements.

- (1) The exact q -coordinates make every literal anchored kernel primitive-supported after the q -sum and the removal of the same-residual direction.
- (2) The canonical zero completion annihilates all $d \geq 2$ diagonal slices and every exact diagonal tail with cutoff $D \geq 1$.
- (3) The anchored coefficient and the complete scalar mixed trace have exact $d = 1$ Abel identities and weighted-plaquette majorants.
- (4) A Banach-valued, anchor-labelled version retains the complete growing anchor family and replaces uniform anchorwise little-oh notation by one global cost inequality.
- (5) Every finitely supported kernel has a unique terminal-rectangle expansion. Mixed variation is precisely the ℓ^1 -mass of its rectangle charges, and diagonal downsampling is contractive.
- (6) Every analytic completion of the same primitive datum gives the same physical sum. Taking the infimum of the full divisor-variation cost over declared completions is an exact sufficient certificate.
- (7) Counterexamples show that zero completion need not be variation-optimal, completion costs can be inflated arbitrarily, primitive support need not cancel, and scalar trace cancellation does not control the erasure operator.

1.2 Logical level

The finite-support identities and completion calculus are L0. The identification of the literal fixed- h_0 kernel with its canonical primitive zero completion is L1: it retains all complete native keys, actual support, literal coefficients, and raw atomic normalization. The paper does not estimate the resulting physical weighted plaquette cost relative to the anchored or trace mass. That missing inequality is the next L2 gate. In particular, eliminating an auxiliary diagonal tail is not a parity breakthrough.

1.3 Organization

Section 2 fixes the physical datum and its support. Section 3 proves tail collapse and the single-slice Abel form. Section 4 gives the canonical rectangle decomposition. Section 5 develops analytic completions. Section 6 treats the order of q , anchor, and trace aggregation. Countermodels appear in section 7. Section 8 states the exact remaining physical gate, and sections 9 and 10 give the adaptive next route and the final claim ledger.

2 The canonical primitive physical datum

We restate the literal input needed from TPC-70 and TPC-71. No estimate is made in this section.

2.1 Rows, anchors, and erasure coordinates

Fix a finite physical scale, one nonzero integer shift h_0 , two distinct active rows $m < n$ satisfying

$$(mn, h_0) = 1,$$

and squarefree row divisors

$$d_m \mid m, \quad d_n \mid n, \quad d_m = ga, \quad d_n = gb, \quad (a, b) = 1.$$

An active row r is a canonical source row with positive raw atomic mass W_r . On the active row space, the atomic-normalized native Gram has the inherited decomposition

$$G_{\text{nat}} = I + R + E,$$

where R retains common-output pairs with equal residual labels and E retains those with unequal residual labels [2]. A complete native key retains the direct side, external block, opposite row, γ -coordinate, physical time, and every other orthogonal output label; equality of times alone is not used as equality of keys.

For the row pair (m, n) , a complete same-residual anchor $\mathbf{a}$ contains a complete key ι_R , its time j_R , and both active row corners

$$mj_R + h_0 = P_m R_m, \quad nj_R + h_0 = P_n R_n, \quad d_m R_m = d_n R_n.$$

At an actual erasure key ι_E , with time j_E , the corresponding targets are

$$mj_E + h_0 = Q_m S_m, \quad nj_E + h_0 = Q_n S_n, \quad d_m S_m \neq d_n S_n.$$

Here P_m, P_n, Q_m, Q_n are the literal largest-prime labels and R_m, R_n, S_m, S_n their cofactors. At a present corner, let $\beta_{\iota, r}$ denote the literal coefficient after every physical mask and profile. The residual-sign-free four-corner amplitude is

$$\mathcal{A}(\iota_E, \iota_R; m, n) = \frac{\overline{\beta_{\iota_E, m}} \beta_{\iota_E, n} \overline{\beta_{\iota_R, n}} \beta_{\iota_R, m}}{W_m W_n}, \quad (1)$$

and is zero if a required corner is absent [4].

Every target and residual label used below lies on the inherited squarefree physical support. In particular, at an erasure key

$$mj_E + h_0 = Q_m S_m, \quad nj_E + h_0 = Q_n S_n,$$

and

$$\mu(mj_E + h_0), \mu(nj_E + h_0), \mu(d_m S_m), \mu(d_n S_n) \in \{-1, 1\}.$$

The analogous nonvanishing conditions are retained at the same-residual anchor. Put

$$H = h_0(n - m).$$

At an actual erasure key, the two target cofactors are S_m, S_n . Their canonical content and primitive coordinates are

$$q = (S_m, S_n), \quad S_m = qu, \quad S_n = qv. \quad (2)$$

The determinant identity of TPC-70 [4] gives

$$q \mid |H|, \quad nuQ_m - mvQ_n = H/q,$$

and the unequal-residual condition is

$$(u, v) \neq (b, a). \quad (3)$$

Every physical coordinate in (2) satisfies

$$(u, v) = 1. \quad (4)$$

Let $k_{a,q}(u, v)$ be the TPC-70 complete literal ray sum of (1) at these coordinates. It contains every physical time, complete native key, largest-prime condition, endpoint, mask, profile, complex phase, and the four-corner normalization $1/(W_m W_n)$. It is zero when the corresponding actual ray is absent. Aggregate the exact contents before taking absolute values:

$$k_a(u, v) = \mathbf{1}_{(u,v) \neq (b,a)} \sum_{q \mid |H|} k_{a,q}(u, v), \quad (u, v) = 1. \quad (5)$$

Different q -slices may cancel at the same primitive coordinate. Accordingly,

$$\text{supp } k_a \subseteq \bigcup_{q \mid |H|} \text{supp } k_{a,q},$$

and equality is not asserted.

2.2 Canonical zero completion

Write

$$\mathcal{P}_2 = \{(u, v) \in \mathbb{N}^2 : (u, v) = 1\}.$$

Definition 2.1 (Support-faithful completion). The canonical physical zero completion of (5) is

$$K_a^0(u, v) = \begin{cases} k_a(u, v), & (u, v) \in \mathcal{P}_2, \\ 0, & (u, v) \notin \mathcal{P}_2, \end{cases} \quad u, v \geq 1. \quad (6)$$

It is zero both at absent primitive rays and at every nonprimitive coordinate.

The superscript 0 denotes zero completion, not the removal of a physical atom. The same-residual direction (b, a) is already absent from the literal erasure datum.

Theorem 2.2 (Literal primitive support). *For every physical anchor $\mathbf{a}$,*

$$\text{supp } K_a^0 \subseteq \mathcal{P}_2. \quad (7)$$

The property is preserved by exact q -aggregation, finite sums over anchors, and multiplication by anchor-dependent scalars.

Proof. Equation (4) follows from the definition $q = (S_m, S_n)$. Thus no actual ray has a nonprimitive coordinate. Definition 2.1 assigns zero to every remaining site. Pointwise sums and scalar multiplication cannot create support outside $\mathcal{P}_2$. $\square$

Remark 2.3 (Canonical means support-faithful). The completion is canonical because it inserts no coefficient at an absent physical coordinate. This does not say that it minimizes mixed variation among all analytic completions. That different optimization problem is treated in section 5.

2.3 The anchored signed coefficient

Let

$$\varepsilon_{\mathfrak{a}} = \mu(a)\mu(b).$$

The frozen-sign theorem of TPC-70 [4] gives the exact anchored scalar coefficient

$$\boxed{\mathfrak{X}_{\mathfrak{a}} = \varepsilon_{\mathfrak{a}} \sum_{u,v \geq 1} \mu(u)\mu(v)K_{\mathfrak{a}}^0(u,v).} \quad (8)$$

The explicit coprimality condition may be omitted because (7) already enforces it. This is a four-corner contribution to $\text{tr}(ER)$; it is not a two-corner entry of the erasure operator.

3 Tail collapse and single-slice Abel summation

For a finitely supported kernel $F : \mathbb{N}^2 \rightarrow \mathbb{C}$, define

$$F^{[d]}(x, y) = F(dx, dy).$$

TPC-71 [5] introduces the exact diagonal tail

$$\mathcal{T}_D(F) = \sum_{d > D} |\mu(d)| \sum_{x,y \geq 1} |\mu(dx)\mu(dy)F(dx, dy)|. \quad (9)$$

Theorem 3.1 (Primitive-support collapse). *If F is supported on $\mathcal{P}_2$, then*

$$F^{[d]} \equiv 0 \quad (d \geq 2). \quad (10)$$

Consequently,

$$\mathcal{T}_D(F) = 0 \quad (D \geq 1). \quad (11)$$

Proof. For $d \geq 2$ and $x, y \geq 1$,

$$\gcd(dx, dy) = d \gcd(x, y) \geq d > 1.$$

Hence $(dx, dy) \notin \mathcal{P}_2$, so $F(dx, dy) = 0$. Every summand in (9) has $d > D \geq 1$, and therefore vanishes. $\square$

Corollary 3.2 (Physical diagonal hierarchy collapses). *For every physical anchor $\mathfrak{a}$ and every $D \geq 1$,*

$$(K_{\mathfrak{a}}^0)^{[d]} \equiv 0 \quad (d \geq 2), \quad \mathcal{T}_D(K_{\mathfrak{a}}^0) = 0.$$

Thus the support-faithful physical branch requires neither a growing diagonal cutoff nor a restricted-Mertens estimate uniform in d .

3.1 Exact weighted-plaquette identity

Set

$$M(x) = \sum_{n \leq x} \mu(n).$$

For a zero-extended finite kernel, write

$$\begin{aligned} \Delta_1 F(x, y) &= F(x, y) - F(x+1, y), \\ \Delta_2 F(x, y) &= F(x, y) - F(x, y+1), \\ \Delta_{12} F(x, y) &= F(x, y) - F(x+1, y) - F(x, y+1) + F(x+1, y+1). \end{aligned}$$

Theorem 3.3 (Finite two-dimensional Abel identity). *If $F : \mathbb{N}^2 \rightarrow \mathbb{C}$ is finitely supported, then*

$$\boxed{\sum_{u,v \geq 1} \mu(u)\mu(v)F(u, v) = \sum_{x,y \geq 1} M(x)M(y)\Delta_{12}F(x, y).} \quad (12)$$

Proof. All sums are finite. Apply one-dimensional summation by parts in u and then in v . $\square$

Remark 3.4. Primitive support is not needed for the Abel identity itself. It is needed at the physical interface: it permits the coprimality mask to be omitted in (8) and makes every $d \geq 2$ slice of the TPC-71 divisor lift vanish separately.

Definition 3.5 (Weighted plaquette cost). For a finitely supported kernel F , define

$$\mathcal{C}_{\square}(F) = \sum_{x,y \geq 1} |M(x)M(y)| |\Delta_{12}F(x,y)|. \quad (13)$$

Corollary 3.6 (Exact anchored certificate). *For every physical anchor,*

$$\mathfrak{X}_{\mathfrak{a}} = \varepsilon_{\mathfrak{a}} \sum_{x,y \geq 1} M(x)M(y) \Delta_{12}K_{\mathfrak{a}}^0(x,y), \quad (14)$$

$$|\mathfrak{X}_{\mathfrak{a}}| \leq \mathcal{C}_{\square}(K_{\mathfrak{a}}^0). \quad (15)$$

Proof. Combine (8) with theorem 3.3, then take absolute values. $\square$

Remark 3.7 (Sharper than a product of maxima). If F is supported in $[1, U] \times [1, V]$, then

$$\mathcal{C}_{\square}(F) \leq \mathfrak{M}(U)\mathfrak{M}(V)\mathcal{V}_{\square}(F),$$

where

$$\mathfrak{M}(U) = \max_{x \leq U} |M(x)|, \quad \mathcal{V}_{\square}(F) = \sum_{x,y} |\Delta_{12}F(x,y)|.$$

The weighted cost retains the actual location of every plaquette and can be strictly smaller. Neither quantity is estimated for the physical kernel here.

Remark 3.8 (What tail has not disappeared). Equation (11) concerns only the auxiliary diagonal tail introduced by coprimality inversion. It says nothing about the geometric short-ray mass, outer physical endpoints, the ultra-tail of the original packet, or downstream boundary reconstruction.

4 Mixed variation as canonical rectangle charge

The remaining canonical cost is concentrated in $\Delta_{12}K^0$. We record its exact geometric meaning.

For $r, s \geq 1$, define the terminal rectangle

$$R_{r,s}(u,v) = \mathbf{1}_{u \leq r} \mathbf{1}_{v \leq s}. \quad (16)$$

It is a rectangle anchored at $(1, 1)$ and terminating at (r, s) .

Theorem 4.1 (Unique terminal-rectangle expansion). *Every finitely supported $F : \mathbb{N}^2 \rightarrow \mathbb{C}$ has the exact expansion*

$$F(u,v) = \sum_{r,s \geq 1} \Delta_{12}F(r,s) R_{r,s}(u,v). \quad (17)$$

The coefficients are unique. Moreover,

$$\Delta_{12}R_{r,s}(x,y) = \mathbf{1}_{(x,y)=(r,s)} \quad (18)$$

and hence

$$\mathcal{V}_{\square}(F) = \sum_{r,s \geq 1} |\Delta_{12}F(r,s)| \quad (19)$$

is exactly the ℓ^1 -mass of the canonical rectangle charges.

Proof. Finite support permits two successive telescoping sums:

$$\begin{aligned} \sum_{r \geq u} \sum_{s \geq v} \Delta_{12} F(r, s) &= \sum_{r \geq u} \sum_{s \geq v} (F(r, s) - F(r+1, s) \\ &\quad - F(r, s+1) + F(r+1, s+1)) \\ &= F(u, v). \end{aligned}$$

This is (17). Direct substitution gives (18). Applying Δ_{12} to any terminal-rectangle expansion therefore recovers every coefficient, proving uniqueness and (19). $\square$

Corollary 4.2 (Weighted rectangle form of the physical coefficient). *For every physical anchor,*

$$\mathfrak{X}_a = \varepsilon_a \sum_{r, s \geq 1} M(r) M(s) \Delta_{12} K_a^0(r, s). \quad (20)$$

Thus the canonical Abel problem is exactly a discrepancy estimate for the literal terminal-rectangle charges against the weight $M(r)M(s)$.

Remark 4.3. Low mixed variation means that the physical datum admits a sparse canonical superposition of large terminal rectangles. Sparse support in the point basis is a different property and can produce maximal rectangle charge.

4.1 Diagonal downsampling

The next identity is useful both for the general TPC-71 certificate and for later analytic completions.

Theorem 4.4 (Downsampling contraction). *Let $F^{[d]}(x, y) = F(dx, dy)$. Then*

$$\Delta_{12} F^{[d]}(x, y) = \sum_{r=dx}^{d(x+1)-1} \sum_{s=dy}^{d(y+1)-1} \Delta_{12} F(r, s). \quad (21)$$

Consequently,

$$\boxed{\mathcal{V}_\square(F^{[d]}) \leq \mathcal{V}_\square(F)}. \quad (22)$$

Proof. Telescoping over the displayed d -by- d block gives

$$\begin{aligned} &\sum_{r=dx}^{d(x+1)-1} \sum_{s=dy}^{d(y+1)-1} \Delta_{12} F(r, s) \\ &= F(dx, dy) - F(d(x+1), dy) - F(dx, d(y+1)) + F(d(x+1), d(y+1)), \end{aligned}$$

which is $\Delta_{12} F^{[d]}(x, y)$. The blocks indexed by $(x, y) \in \mathbb{N}^2$ are disjoint. Apply the triangle inequality and sum. $\square$

Remark 4.5. For the canonical physical completion, [theorem 3.1](#) is much stronger: $F^{[d]}$ is identically zero for $d \geq 2$. The contraction theorem is needed when a nonprimitive analytic completion is deliberately introduced.

4.2 Point and rectangle extremes

For $p, q \geq 1$, write

$$e_{p,q}(u, v) = \mathbf{1}_{(u,v)=(p,q)}.$$

If $p, q > 1$, a point mass $ce_{p,q}$ has four nonzero rectangle charges and

$$\mathcal{V}_\square(ce_{p,q}) = 4|c|.$$

By contrast, a constant terminal rectangle $cR_{U,V}$ has one charge and

$$\mathcal{V}_\square(cR_{U,V}) = |c|.$$

These are exact identities, not asymptotic estimates. They explain why filling nonprimitive holes may trade pointwise sparsity for rectangular coherence.

5 Analytic completions and the coupled certificate gauge

The physical datum is prescribed only on the primitive lattice, including zeros at absent primitive rays. Values at nonprimitive coordinates do not change the physical sum. They can nevertheless change every term of a triangle-inequality certificate.

5.1 Declared finite completions

Fix $U, V \in \mathbb{N}$ and a rectangle

$$B_{U,V} = [1, U] \times [1, V]$$

containing the primitive physical datum

$$k : \mathcal{P}_2 \cap B_{U,V} \longrightarrow \mathbb{C}.$$

Definition 5.1 (Completion class). Let $\text{Ext}_{U,V}(k)$ be the affine space of kernels $F : \mathbb{N}^2 \rightarrow \mathbb{C}$ such that

$$\text{supp } F \subseteq B_{U,V}, \quad F(u, v) = k(u, v) \quad ((u, v) \in \mathcal{P}_2 \cap B_{U,V}).$$

The values at nonprimitive sites are free analytic variables. The support-faithful zero completion

$$K^0(u, v) = \begin{cases} k(u, v), & (u, v) \in \mathcal{P}_2 \cap B_{U,V}, \\ 0, & \text{otherwise} \end{cases}$$

belongs to this class.

Define the primitive signed functional

$$\mathcal{S}_{\text{prim}}(k) = \sum_{\substack{u \leq U, v \leq V \\ (u,v)=1}} \mu(u)\mu(v)k(u, v). \quad (23)$$

Theorem 5.2 (Completion invariance). *For every $F \in \text{Ext}_{U,V}(k)$,*

$$\boxed{\mathcal{S}_{\text{prim}}(k) = \sum_{\substack{u, v \geq 1 \\ (u,v)=1}} \mu(u)\mu(v)F(u, v).} \quad (24)$$

Thus the physical functional is constant on the complete affine class $\text{Ext}_{U,V}(k)$.

Proof. The summation in (24) samples only primitive coordinates, where every member of the class equals k . $\square$

5.2 The full completion cost

For $d \geq 1$, put

$$M_d(x) = \sum_{\substack{n \leq x \\ (n,d)=1}} \mu(n), \quad \mathfrak{M}_d(T) = \max_{1 \leq n \leq \lfloor T \rfloor} |M_d(n)|, \quad F^{[d]}(x, y) = F(dx, dy).$$

We set $\mathfrak{M}_d(T) = 0$ for $T < 1$. For $D \geq 1$, define the exact weighted completion cost

$$\begin{aligned} \mathcal{C}_D(F) &:= \sum_{d \leq D} |\mu(d)| \sum_{x, y \geq 1} |M_d(x)M_d(y)| |\Delta_{12} F^{[d]}(x, y)| \\ &\quad + \mathcal{T}_D(F), \end{aligned} \quad (25)$$

where $\mathcal{T}_D$ is the exact tail in (9).

Theorem 5.3 (Completion-gauge certificate). *For every declared completion class and every $D \geq 1$,*

$$|\mathcal{S}_{\text{prim}}(k)| \leq \inf_{F \in \text{Ext}_{U,V}(k)} \mathcal{C}_D(F). \quad (26)$$

For the canonical zero completion,

$$\mathcal{C}_D(K^0) = \mathcal{C}_{\square}(K^0) \quad (D \geq 1). \quad (27)$$

Proof. Fix $F \in \text{Ext}_{U,V}(k)$. By [theorem 5.2](#), its primitive sum is the physical functional. The exact TPC-71 diagonal divisor lift [\[5\]](#) followed by two-dimensional Abel summation gives

$$\mathcal{S}_{\text{prim}}(k) = \sum_{d \leq D} \mu(d) \sum_{x,y \geq 1} M_d(x) M_d(y) \Delta_{12} F^{[d]}(x, y) + \mathcal{R}_D(F), \quad |\mathcal{R}_D(F)| \leq \mathcal{T}_D(F).$$

Taking absolute values gives

$$|\mathcal{S}_{\text{prim}}(k)| \leq \mathcal{C}_D(F).$$

Take the infimum. For K^0 , all $d \geq 2$ slices and the tail vanish by [theorem 3.1](#); the $d = 1$ term is exactly [\(13\)](#). $\square$

Remark 5.4 (Infimum versus a physical construction). The infimum is an exact finite-dimensional analytic benchmark. To use a completion in a uniform physical theorem, one must construct it before any later extremizing coefficient vector is selected and control its cost uniformly in all literal labels. An optimizer chosen after seeing the operator extremizer changes the quantifier. Moreover, nonprimitive values are analytic fill-ins and are never added to the physical reference mass.

Remark 5.5 (No separate monotonicity). The $d = 1$ variation and the higher-diagonal tail can move in opposite directions when a completion changes. Neither term is an invariant of the primitive datum. Only the original physical functional and a proved bound for the complete cost are invariant conclusions.

5.3 A coarse completion norm

The weighted cost is the full location-sensitive deterministic majorant. A coarser but sometimes convenient completion cost is

$$\mathcal{E}_D(F) := \sum_{d \leq D} |\mu(d)| \mathfrak{M}_d(U/d) \mathfrak{M}_d(V/d) \mathcal{V}_{\square}(F^{[d]}) + \mathcal{T}_D(F). \quad (28)$$

Then $\mathcal{C}_D(F) \leq \mathcal{E}_D(F)$. The two displayed pieces in [\(28\)](#) must be evaluated for the same completion; minimizing them independently would splice incompatible analytic objects.

6 Aggregation over contents, anchors, and row pairs

All index sets at a finite physical scale are finite. We may therefore aggregate exact scalar kernels before applying absolute values.

6.1 The q -sum

Let $K_{a,q}^0$ be the canonical zero completion of the q -slice, with the excluded direction removed. Linearity gives

$$K_a^0 = \sum_{q||H} K_{a,q}^0, \quad \Delta_{12} K_a^0 = \sum_{q||H} \Delta_{12} K_{a,q}^0. \quad (29)$$

Proposition 6.1 (Post-aggregation sharpness). *One has*

$$\mathcal{C}_{\square}(K_a^0) \leq \sum_{q||H} \mathcal{C}_{\square}(K_{a,q}^0), \quad (30)$$

$$\mathcal{V}_{\square}(K_a^0) \leq \sum_{q||H} \mathcal{V}_{\square}(K_{a,q}^0). \quad (31)$$

Both inequalities can be strict in the ambient class of finite primitive-supported kernels, including the case in which the left side is zero and the right side is positive. This does not assert that every literal physical family realizes such exact slice cancellation.

Proof. Apply the triangle inequality pointwise to (29), first with weights $|M(x)M(y)|$, then without them. Ambient strictness follows by taking two nonzero slices F and $-F$. $\square$

Thus the exact q -sum precedes absolute values. The divisor bound $\tau(|H|) = |H|^{o(1)}$ controls the number of slices only; it supplies no signed cancellation.

6.2 Scalar trace kernel

For each row pair $m < n$, let $\mathcal{R}_{m,n}$ be its complete set of actual same-residual anchors. Extend every finite kernel by zero to $\mathbb{N}^2$ and define

$$K_{\text{tr}}^0(u, v) = \sum_{m < n} \sum_{\mathbf{a} \in \mathcal{R}_{m,n}} \varepsilon_{\mathbf{a}} K_{\mathbf{a}}^0(u, v). \quad (32)$$

Although the coordinate labels originate from different row pairs, the right side is only a scalar reindexing of a finite sum. It is not an operator identification.

Theorem 6.2 (Global trace Abel summation). *The complete first mixed trace satisfies*

$$\text{tr}(ER) = 2 \operatorname{Re} \sum_{u, v \geq 1} \mu(u) \mu(v) K_{\text{tr}}^0(u, v) \quad (33)$$

$$= 2 \operatorname{Re} \sum_{x, y \geq 1} M(x) M(y) \Delta_{12} K_{\text{tr}}^0(x, y). \quad (34)$$

Consequently,

$$|\text{tr}(ER)| \leq 2 \mathcal{C}_{\square}(K_{\text{tr}}^0) \leq 2 \sum_{m < n} \sum_{\mathbf{a} \in \mathcal{R}_{m,n}} \mathcal{C}_{\square}(K_{\mathbf{a}}^0). \quad (35)$$

Proof. TPC-70 [4] gives

$$\text{tr}(ER) = 2 \operatorname{Re} \sum_{m < n} \sum_{\mathbf{a} \in \mathcal{R}_{m,n}} \mathfrak{X}_{\mathbf{a}}.$$

Insert (8) and interchange finite sums to obtain (33). The kernel in (32) remains primitive-supported, so [theorem 3.3](#) gives (34). Apply the triangle inequality once after full scalar aggregation, or before anchor aggregation for the second bound. $\square$

6.3 Anchor-safe labelled form

Let $\mathcal{I}_X$ be the finite set of all triples $(m, n, \mathbf{a})$, and define the $\ell^1(\mathcal{I}_X)$ -valued kernel

$$\mathbf{K}^0(u, v) = (\varepsilon_{\mathbf{a}} K_{\mathbf{a}}^0(u, v))_{(m, n, \mathbf{a}) \in \mathcal{I}_X}. \quad (36)$$

Proposition 6.3 (Anchor-safe global certificate). *One has the exact vector identity*

$$\sum_{u, v} \mu(u) \mu(v) \mathbf{K}^0(u, v) = (\mathfrak{X}_{\mathbf{a}})_{(m, n, \mathbf{a}) \in \mathcal{I}_X}$$

and

$$\sum_{m < n} \sum_{\mathbf{a} \in \mathcal{R}_{m,n}} |\mathfrak{X}_{\mathbf{a}}| \leq \sum_{m < n} \sum_{\mathbf{a} \in \mathcal{R}_{m,n}} \mathcal{C}_{\square}(K_{\mathbf{a}}^0). \quad (37)$$

Proof. Apply the Banach-valued two-dimensional Abel identity in $\ell^1(\mathcal{I}_X)$. The norm of the mixed difference is the sum of the coordinatewise absolute values, giving the right side of (37). $\square$

Remark 6.4 (Uniform anchors are not required). A sufficient trace-level theorem may prove that the total cost on all anchors is small relative to the declared total post- q -aggregation mass. Individual bad anchors are allowed if their aggregate cost is negligible. This is weaker than a uniform anchorwise estimate and stronger than fixed-anchor little-oh notation with no domination.

Remark 6.5 (Still not an operator theorem). Cancellation in K_{tr}^0 can occur across anchors and row pairs. Neither (35) nor (37) bounds a two-corner entry, row star, Perron radius, or operator norm. A direct-entry or row-local bridge remains necessary.

7 Countermodels and non-substitution laws

The following finite examples separate the conclusions proved above from the estimates which remain open.

7.1 Zero completion need not minimize variation

Fix $U, V \geq 3$ and prescribe the primitive datum

$$k(u, v) = 1 \quad ((u, v) = 1, \ u \leq U, \ v \leq V).$$

Its canonical zero completion is

$$K^0(u, v) = \mathbf{1}_{u \leq U, v \leq V} \mathbf{1}_{(u, v) = 1}.$$

The constant rectangle

$$F(u, v) = \mathbf{1}_{u \leq U, v \leq V}$$

belongs to the same completion class. By [theorem 4.1](#),

$$\mathcal{V}_{\square}(F) = 1.$$

The zero completion is much rougher. Put $L = \min(U, V)$. For every $2 \leq n \leq L - 1$,

$$K^0(n, n) = K^0(n + 1, n + 1) = 0, \quad K^0(n + 1, n) = K^0(n, n + 1) = 1,$$

and hence

$$\Delta_{12}K^0(n, n) = -2.$$

Therefore

$$\mathcal{V}_{\square}(K^0) \geq 2(L - 2). \tag{38}$$

The completion F reduces the $d = 1$ variation, but $F^{[d]}$ is nonzero for many d and its diagonal tail need not vanish. This is the basic variation-tail trade-off.

Remark 7.1. The example does not show that the full cost $\mathcal{C}_D(F)$ is smaller than $\mathcal{C}_{\square}(K^0)$ at every scale. It shows only that zero completion is not variation-minimal, which is the precise claim needed to prevent a false optimization statement.

7.2 Completion cost can be inflated arbitrarily

Take the zero primitive datum $k = 0$, a prime $p \leq \min(U, V)$, and

$$F_A = Ae_{p,p}.$$

The point (p, p) is nonprimitive, so $F_A \in \text{Ext}_{U,V}(0)$ and the physical functional is zero. For $D = 1$, the exact tail contains the term $d = p, x = y = 1$, giving

$$\mathcal{T}_1(F_A) \geq |A|.$$

Thus

$$\mathcal{C}_1(F_A) \geq |A|$$

can be arbitrarily large while the physical datum and physical sum remain zero. A large cost for an arbitrary analytic completion is not a physical obstruction.

7.3 Primitive support and tail freedom do not cancel

Let $(u_0, v_0) = 1$ and

$$K^0 = ce_{u_0, v_0}.$$

It is primitive-supported and has zero diagonal tail, but

$$\left| \sum_{u,v} \mu(u)\mu(v)K^0(u,v) \right| = |c| |\mu(u_0)\mu(v_0)|.$$

When u_0, v_0 are squarefree this equals the full ℓ^1 -mass $|c|$. Tail freedom is therefore an exact bookkeeping improvement, not an arithmetic cancellation theorem.

7.4 Slice and anchor cancellation are different

In the ambient finite-kernel class, two formal q -slices F and $-F$ have zero post- q -aggregation kernel although the sum of the two separate costs is positive. Likewise, an abstract anchor-compatible model may have

$$\varepsilon_{a_1} K_{a_1}^0 = F, \quad \varepsilon_{a_2} K_{a_2}^0 = -F.$$

Then $K_{\text{tr}}^0 = 0$ and the scalar trace certificate vanishes, while the anchor-labelled ℓ^1 cost is twice the cost of F . Consequently:

$$\begin{aligned} \text{separate } q\text{-cost} &\not\equiv \text{post-}q\text{-aggregation cost,} \\ \text{scalar trace cancellation} &\not\equiv \text{anchorwise or row-local control.} \end{aligned}$$

7.5 A vanishing mixed carrier does not control the erasure operator

The abstract example

$$R = 0, \quad E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

has

$$\text{tr}(ER) = 0, \quad \|E\| = 1.$$

In this abstract model the mixed carrier ER vanishes identically, while the erasure operator is nonzero. No scalar completion theorem can remove the need for a genuine two-corner, row-local, or complete-moment operator carrier.

7.6 Quantifier-adaptive completion is invalid

For each fixed primitive datum, finite-dimensional optimization can choose a completion with small cost. A coefficient-uniform operator theorem requires a single declared completion rule valid for the entire physical coefficient class. Choosing a different completion after the maximizing coefficient vector or row is known reverses the max-min order. The pointwise infimum in (26) is therefore an analytic benchmark; it is not automatically a uniform L2 theorem.

8 The reduced physical gate

The canonical physical branch now has no diagonal-cutoff choice. Its remaining trace-level problem is a weighted plaquette estimate. From this point, let $X \rightarrow \infty$ through one declared physical family. The active rows, anchors, kernels, and reference scales all depend on X ; the index is suppressed, and every $o(\cdot)$ statement is with respect to this limit.

8.1 Reference masses

For one anchor define the exact post- q -aggregation mass

$$\mathfrak{W}_a = \sum_{u,v \geq 1} |K_a^0(u,v)|. \quad (39)$$

Absolute values occur after the exact q -sum. This mass is neither the sum of pre-aggregation ray masses nor automatically the raw global Gram mass. Let $\mathfrak{A}_{\text{tr}} > 0$ be one explicitly declared physical reference scale for the complete first mixed trace.

Proposition 8.1 (Canonical physical closure criterion). *If*

$$\sum_{m < n} \sum_{a \in \mathcal{R}_{m,n}} \mathcal{C}_{\square}(K_a^0) = o(\mathfrak{A}_{\text{tr}}), \quad (40)$$

then

$$\text{tr}(ER) = o(\mathfrak{A}_{\text{tr}}). \quad (41)$$

The conclusion remains valid if the left side of (40) is replaced by the sharper $\mathcal{C}_{\square}(K_{\text{tr}}^0)$.

Proof. Use (35). □

This is an exact sufficient condition, not a theorem that the physical cost is small. It allows a nonuniform anchor family because the quantifier is global and the total cost is explicit.

8.2 Completion route

For each anchor choose a declared analytic completion

$$F_a \in \text{Ext}_{U_a, V_a}(k_a)$$

before any later operator extremizer is selected. The general sufficient condition is

$$\sum_{m < n} \sum_{a \in \mathcal{R}_{m,n}} \mathcal{C}_{D_a}(F_a) = o(\mathfrak{A}_{\text{tr}}). \quad (42)$$

Here variation and diagonal tail refer to the same completion. A future physical construction may, for example, move a common coordinate factor into an existing q -label and fill only those nonprimitive sites compatible with literal content data. An arbitrary smooth interpolation is an L0 analytic device until its coefficient norm, support, and quantifier dependence are controlled.

8.3 Exact stop rules

The route has the following logically different outcomes.

- (G1) If (40) fails by a proved lower obstruction on a nonnegligible physical family, only the canonical tail-free certificate is closed.
- (G2) To close the full completion route, one needs a lower obstruction for every declared admissible completion rule, or for the infimum in (26), at the same physical mass scale.
- (G3) Failure of an ambient tail estimate closes only that ambient specialization. The auxiliary diagonal tail of the canonical zero extension is already zero.
- (G4) A large zero-completion variation does not prove that the original signed sum is large. Literal complex phases, a structured completion, or a different signed sector may still cancel.
- (G5) A completed estimate for $\text{tr}(ER)$ remains trace-level. It does not imply an entrywise erasure bound, a row star, a Perron estimate, or an operator norm.
- (G6) The geometric short-ray mass, physical endpoint cells, represented lower frame, missing-mass comparison, nuisance shorting, and final boundary reconstruction are untouched.

8.4 Exponent interpretation

The classical prime number theorem gives $M(x) = o(x)$ [1]. It does not by itself provide the fixed polynomial saving required by the strict endpoint ledger. For a coarse anchorwise estimate, one would need quantitative control of

$$\mathfrak{M}(U_a)\mathfrak{M}(V_a)\mathcal{V}_\square(K_a^0)$$

relative to the literal reference mass. Calling $M(x) = o(x)$ an endpoint power saving would be incorrect.

9 Adaptive route from TPC–73 to TPC–82

TPC–72 separated four live tasks: physical primitive-plane variation, a genuine operator carrier, represented-frame closure, and final shorted reassembly. The present paper removes the auxiliary diagonal tail for the support-faithful physical zero extension and turns the first task into a completion and weighted-plaquette problem. The next block will alternate among the independent branches rather than refine one branch indefinitely.

9.1 Provisional sequence

Paper	Door	Falsifiable target
TPC–73	Primitive support	Tail collapse, completion invariance, and the exact weighted plaquette gate.
TPC–74	Direct two-corner carrier	Reindex the literal erasure entry itself on primitive cofactor coordinates, or prove that a required physical sector lacks such a canonical reindexing.
TPC–75	Completion/coherence census	Construct a q -compatible completion with controlled full cost, or give an actual-carrier lower obstruction for the completion certificate.
TPC–76	Row-local observability	Transfer anchored data to observable row stars with a quantified inverse, while isolating orphan erasure edges.
TPC–77	Terminal activation	Use literal moment-to-capacity estimates in a declared distinguished coefficient scope; do not replace them by a uniform worst-row claim.
TPC–78	Represented determinant capacity	Convert actual good-component determinant mass into a distinguished lower frame, or close that certificate on an actual bad-energy family.
TPC–79	Operator aggregation	Apply pair/star/Perron or complete moments only to the literal carrier established in TPC–74–TPC–76.
TPC–80	Nuisance principal angle	Express shorting loss through the angle between represented output and the missing image of the represented kernel.
TPC–81	Compatible merge	Test missing mass, represented frame, nuisance, boundaries, and all fixed-power losses in one carrier and quantifier scope.
TPC–82	Ten-paper audit	Classify exact advances, actual obstructions, unresolved L2 inputs, fallbacks, and the strict endpoint status.

9.2 Dynamic branching rules

The numbering describes dependencies, not guaranteed positive outcomes.

- (R1) If TPC–75 proves that every admissible completion has excessive physical cost, the Abel branch is recorded as a negative certificate theorem and the signed effort moves to direct entries, literal phases, or higher one-erasure polygons.

- (R2) If the direct-entry carrier closes in TPC–74, it has priority over anchor observability because it also sees orphan erasure edges. Otherwise TPC–76 retains the exact observable/orphan decomposition.
- (R3) A positive trace theorem never substitutes for TPC–79. Operator aggregation begins only after a two-corner, row-local, or complete-moment carrier exists.
- (R4) The activation and represented papers proceed independently of the primitive-plane branch. Failure of one branch does not supply a hypothesis on the other.
- (R5) TPC–81 attempts a merge only if all inputs use literal coefficients, the same fixed h_0 , raw physical normalization, actual support, and one common uniform or distinguished coefficient scope.
- (R6) Any actual family forcing total fixed-power loss at least $1/400$ closes the present endpoint implication, including equality.

At every stage, success means a literal exponent improvement, closure of a necessary interface, or an exact actual-carrier obstruction. A numerical trend, a smooth surrogate, a fixed anchor, or an averaged shift is not reported as fixed-shift arithmetic progress.

10 Claim ledger and conclusion

10.1 Exact conclusions

The paper proves the following.

- (1) The literal q -aggregated physical kernel has canonical support on the primitive lattice.
- (2) Its support-faithful zero completion has no $d \geq 2$ diagonal slice and no exact diagonal tail for any $D \geq 1$.
- (3) The anchored coefficient, scalar global trace kernel, and anchor-labelled kernel have exact ordinary $d = 1$ Abel identities.
- (4) The weighted plaquette cost gives the coefficient-faithful, location-sensitive canonical certificate and can be aggregated before or after retaining anchor labels.
- (5) Mixed variation equals the ℓ^1 -mass of the unique terminal-rectangle coefficients, and diagonal downsampling is contractive.
- (6) The physical signed functional is invariant under analytic completion at nonprimitive sites.
- (7) The complete divisor-variation cost of one completion is a valid certificate; its infimum is an exact analytic benchmark.
- (8) Canonical zero completion is tail-free but need not be variation-optimal.

10.2 Statements not proved

The paper does not prove:

- that $\mathcal{C}_{\square}(K_{\mathfrak{a}}^0)$ is small relative to $\mathfrak{W}_{\mathfrak{a}}$;
- that the global cost in (40) is $o(\mathfrak{A}_{\text{tr}})$;
- a uniformly controlled low-cost analytic completion for the literal physical family;
- a fixed-power Möbius saving for the physical first mixed trace;
- a direct two-corner representation of $E_{m,n}$;

- an entrywise, row-star, Perron, or operator-norm saving;
- a represented lower frame, missing-mass estimate, or nuisance-angle estimate;
- closure of geometric short rays, physical endpoints, or the final packet tail;
- the strict endpoint inequality for one complete chain;
- any result using a property special to $h_0 = 2$; or
- a parity-breaking, prime-pair, or twin-prime theorem.

10.3 Audit levels

Result	Scope	Level
Rectangle decomposition, downsampling, completion gauge	All finite plane kernels and declared completion classes	L0
Primitive support and diagonal-tail collapse	Literal q -canonical fixed- h_0 anchored kernel	L1
Global scalar and anchor-labelled trace identities	Complete actual anchor family, trace scope only	L1
Mass-normalized weighted plaquette saving	Required for the actual growing physical family	Open L2
Trace-to-operator promotion	Requires a new two-corner, row-local, or moment carrier	Open L2

10.4 Conclusion

The auxiliary diagonal tail isolated in TPC–71 and carried as an open certificate gate in TPC–72 is not present for the canonical primitive zero completion. That closure is exact and useful: it removes a cutoff, all growing- d uniformity, and one possible source of endpoint loss. The result does not make the arithmetic easy. It concentrates the canonical route in one literal object,

$$\sum_{x,y} |M(x)M(y)| |\Delta_{12} K_a^0(x, y)|,$$

and exposes analytic completion as a coupled variation-tail optimization, not as free smoothing. A natural next question is whether primitive coordinates can be attached directly to the two-corner erasure entry. The plane and represented branches remain available in parallel if that door closes.

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